REGRESSION ANALYSIS OF DIABETES INDICATORS IN PIMA INDIAN WOMEN

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GROUP 1

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Introduction:

Diabetes is a chronic disease influenced by both genetic and environmental factors. Within the US, approximately 1 in 10 individuals have diabetes with 90% of cases being type 2 (CDC, 2023). The Pima Indians Diabetes Dataset is a renowned resource in medical research, particularly in studying diabetes. It consists of diagnostic measurements from female patients of Pima Indian heritage, a population with a high incidence of diabetes. The primary research question was to identify which variables are significantly associated with the likelihood of a diabetes diagnosis among Pima Indian women. Our research explores the variables within the Pima dataset that are most strongly associated with a diagnosis of diabetes within 5 years, and how these variables individually and collectively contribute to predicting the binary outcome of a diabetes diagnosis.

Dataset Collection:

The data of interest for this project, pulled from the Pima longitudinal study, is the Pima Indians Diabetes Database (Smitt et al., 1988. Originally, this dataset was collected to validate the predictive power of the ADAP machine learning model by Smith et al., (1988). Specifically, researchers were interested in predicting the likelihood of diabetes diagnosis within 5 years. The diagnostic criteria at the time of the paper were defined as a measure of 2-hour post-load plasma glucose at 200 (mg/dl). The Pima Indians Diabetes dataset was meticulously collected from Pima Indian women residing near Phoenix, Arizona. The participants were systematically chosen, focusing on females aged 21 years and older. This method ensures a representative sample of the population known for its high diabetes prevalence, providing a comprehensive overview for analysis.

Variables:

The dataset encompasses 768 records related to diabetes with several critical variables necessary for diabetes research. These include the number of pregnancies, plasma glucose concentration, diastolic blood pressure, triceps skin fold thickness, 2-hour serum insulin, body mass index (BMI), diabetes pedigree function, age, and diabetic outcome (0 for non-diabetic, 1 for diabetic). Each of these variables plays a crucial role in diagnosing and understanding the dynamics of diabetes within the community.

Summaries:

The data reveals an average age of 33 years, a mean BMI of 31.99 suggesting an overweight or obese population, and an average glucose level of 121. Notably, insulin levels show considerable variability, with an average of 80 mu U/ml.

Initial Observations:

There are no missing values in the dataset, but certain physiological measures like glucose and blood pressure show null values, possibly indicating placeholders or errors. The data necessitates further examination of these null values and exploring the relationships between features and diabetic outcomes.

Explaining the statistical methods:

Data Cleaning:

Our initial focus was on preparing a clean, accurate data set to ensure the validity of our findings. We removed any entries with missing information to avoid the potential bias that incomplete data might introduce. We then transformed the 'diabetes' column into a straightforward binary indicator that would serve as a clear dependent variable for subsequent analysis. This binary outcome—indicating the presence or absence of diabetes—was essential for applying logistic regression techniques later in our analysis.

Exploratory Data Analysis (EDA):

Before hypothesis testing, we conducted an exploratory data analysis to uncover patterns, spot anomalies, identify trends, and test assumptions. This included visualizing the distribution of each variable using histograms and boxplots to assess the data's spread and central tendency, as well as identify outliers. We used scatter plots to examine potential relationships between variables and employed correlation matrices to quantify the strength of these associations.

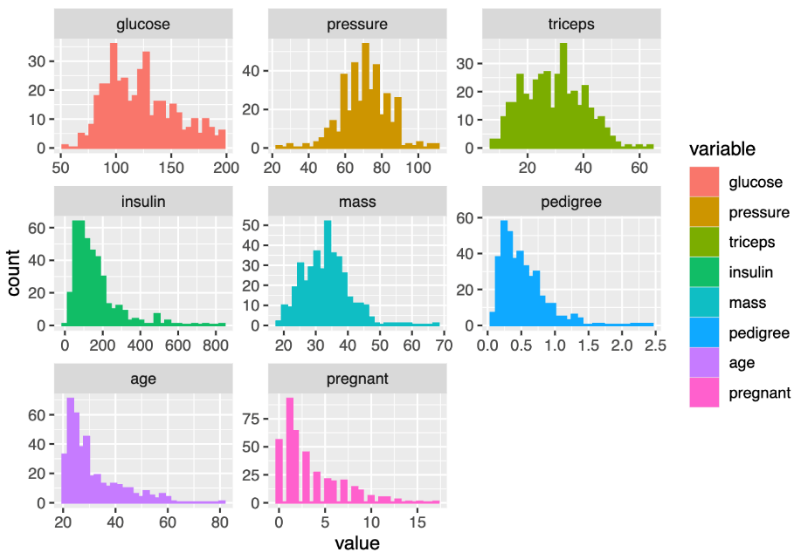

Fig: 1

Rationale for Statistical Methods:

The choice of EDA techniques was guided by our intent to understand the underlying structure of the dataset before applying more complex models. The rationale for using logistic regression involves its ability to provide odds ratios, which offer interpretable insights into how much each predictor increases or decreases the odds of diabetes. This is crucial for understanding which factors are most impactful, allowing healthcare professionals to target these areas.

Appropriateness of Statistical Methods:

Given the binary nature of the outcome (presence or absence of diabetes), using logistic regression is appropriate because it allows for the estimation of probability based on predictor variables. Logistic regression is particularly suited for cases where you need to understand the impact of several independent variables on a binary outcome.

Limitations:

The statistical methods used in the Pima Indians Diabetes Dataset study, primarily logistic regression, assume linear relationships and independent observations, which may not hold in community-based settings where interdependencies exist. Utilizing multiple predictors could lead to model overfitting, especially if the complexity is unsupported by the sample size, affecting the model's generalizability. Additionally, the handling of missing data by excluding incomplete records could introduce bias, particularly if the missingness is not random, potentially skewing the study results.

Explaining the findings:

Question Addressed:

The primary research question was to identify which variables are significantly associated with the likelihood of a diabetes diagnosis among Pima Indian women.

Correlation Analysis

The correlation analysis sought to understand the pairwise relationships between continuous variables. The Pearson correlation coefficients quantify the degree to which two variables linearly relate to each other. Logistic regression does not require the same assumptions of linearity and normality in residuals as linear regression. It instead requires a lack of multi-collinearity or strong associations within the data set. Using correlation analysis allows us to determine if this assumption is met.

From the results of the correlation analysis, we found that there is a moderate positive correlation observed between glucose and insulin (r = 0.58), indicating that as blood glucose levels rise, insulin levels tend to increase correspondingly, which is biologically expected given insulin's role in regulating blood glucose. A similar moderate positive correlation between triceps skinfold thickness and BMI (r = 0.66) is observed, suggesting that as body fat increases, overall body mass also increases. This reflects the known link between adiposity and overall body mass.
With only moderate correlations between the independent variables of this dataset, we can be satisfied that there is little multicollinearity in the data.

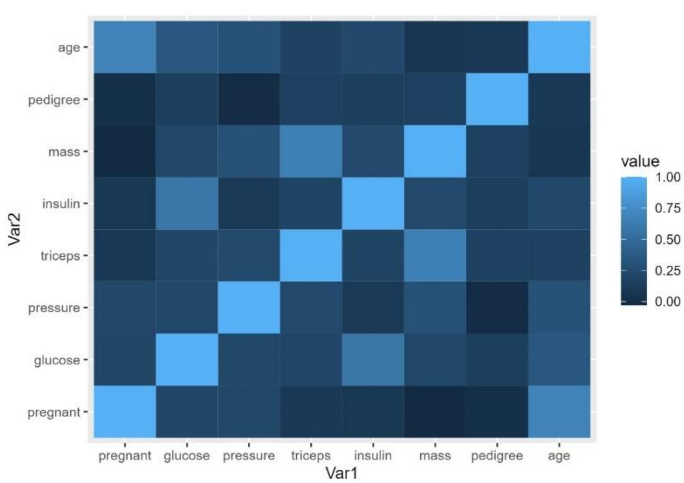

Fig: 2

By applying Spearman correlation alongside Pearson, we aim to detect any non-linear relationships or ordinal associations between variables that Pearson might miss due to its assumptions. The similarity in results from both Pearson and Spearman correlations implies that there are likely no hidden non-linear trends between your variables that could complicate linear analyses like logistic regression. It also suggests that the data does not suffer significantly from non-normal distributions, at least in terms of how these distributions impact the relationships between variables.

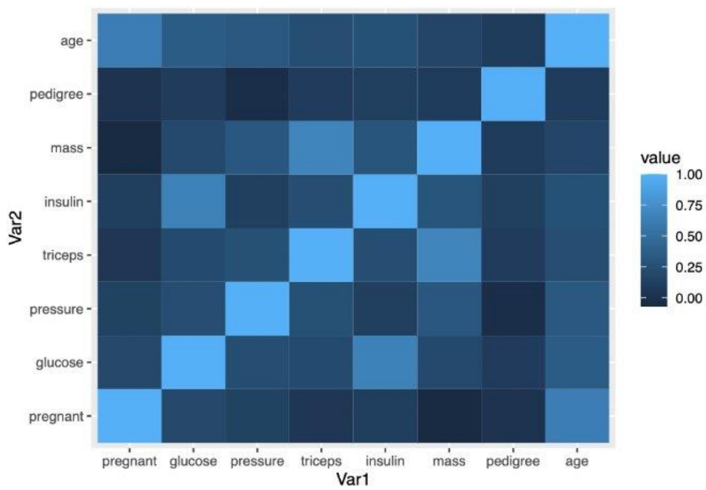

Fig: 3

Univariate Logistic Regression:

With a binary outcome variable, we cannot use correlation to evaluate the relationship between the predictor variables and the outcome. Instead, simple logistic regression can illustrate the strength and direction of the relationship between each variable and diabetes outcome. This approach modeled the log odds of diabetes as a function of one predictor at a time. The analysis quantifies how each variable individually affects the likelihood of developing diabetes. These were the significant results found after running this analysis.

- Glucose: A unit increase in glucose levels resulted in a 4.2% increase in the odds of diabetes (OR ≈ 1.042), showcasing glucose as a critical factor in diabetes risk.
- BMI (Mass): A unit increase in BMI was associated with an 8.6% increase in diabetes odds (OR ≈ 1.086), highlighting the importance of body mass in the condition's development.
- Diabetes Pedigree Function (Pedigree): A significant predictor with a strong effect size (OR ≈ 2.582), indicating a profound genetic contribution to diabetes risk.
- Age: With each additional year of age, the odds of diabetes increased by 7.4% (OR ≈ 1.077), underscoring age as a risk factor.

$$y = \beta_0 + \beta_1 x + \epsilon$$

Multivariate Logistic Regression

The multivariate logistic regression uses multiple independent variables to model the log odds of the dependent variable, in this case, the presence of diabetes. This method adjusts for confounders and can reveal the unique contribution of each variable while holding others constant. This is evidenced by the change in significance for pregnancy between the univariate and multivariate models. While pregnancy alone had a significance level below .05 in the univariate model, when paired with age in the multivariate model, we saw its significance diminish. This indicates a likely confounding effect of age on pregnancy. Similarly, blood pressure, skin fold, and insulin also exhibited changes in significant levels when considered alongside other data. The multivariate model identified glucose, age, pedigree, and BMI as significant predictors. These were chosen because they demonstrated a consistent and significant relationship with the outcome, even when accounting for other variables in the model.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon$$

Model Comparison using AIC:

The Akaike Information Criterion (AIC) was utilized to compare different models' goodness of fit while penalizing for the number of predictors used, aiming for a model that provides the best balance between complexity and explanatory power.

The "Strong vars" model achieved the lowest AIC score, suggesting it has the most appropriate number of parameters providing significant information without overfitting the data.

$$\text{k} = \text{number of parameters} \text{LL} = \text{log-likelihood} AIC = 2 * (k - LL)$$

Best Fit Model Analysis - Log Odds Ratios:

From the best-fit model, log odds ratios were extracted, translating the coefficients into multiplicative effects on the odds of diabetes for each unit increase in the predictors.

Glucose, age, pedigree, and BMI showed significant positive log odds ratios, indicating their incremental effect on diabetes risk. Notably, glucose had an odds ratio close to 1.037, age had 1.054, pedigree had 2.966 and BMI had 1.081, meaning that these variables are predictive of diabetes risk with varying magnitudes of effect, with pedigree showing a slightly stronger association per unit increase.

The results collectively underscore the significance of both modifiable factors (like glucose levels and BMI) and non-modifiable factors (like age and genetic pedigree) in predicting diabetes, providing a strong foundation for targeted preventative healthcare strategies.

Table 5:

	Dependent variable:
	outcome
	Strong vars
glucose	1.037 (1.027, 1.047)
	t = 7.264
	p = 0.000
age	1.054 (1.028, 1.083)
	t = 3.945
	p = 0.0001
mass	1.077 (1.036, 1.122)
	t = 3.673
	p = 0.0003
pedigree	2.966 (1.327, 6.871)
	t = 2.592
	p = 0.010
Constant	0.00004 (0.00000, 0.0003)
	t = −9.342
	p = 0.000
Observations	392
Log Likelihood	−173.617
Akaike Inf. Crit.	357.235
Note:	*p<0.1; **p<0.05; ***p<0.01

Fig: 4

Conclusion:

Through meticulous analysis of the Pima Indians Diabetes Dataset, our study has identified key factors—plasma glucose levels, body mass index (BMI), age, and genetic predisposition—that significantly influence the likelihood of diabetes diagnosis within five years among Pima Indian women. The rigorous application of both univariate and multivariate logistic regression models established that these variables contribute significantly both individually and collectively to diabetes risk which would reject the null hypothesis. Our findings reinforce the critical importance of monitoring these variables, with glucose levels showing a particularly strong correlation, echoing established medical insights.

Despite the robust analytical framework employed, the study acknowledges limitations such as the exclusion of records with missing data and assumptions of linearity and independence in logistic regression, which may impact the generalizability of our conclusions. Nevertheless, the insights garnered hold substantial implications for targeted interventions and preventive healthcare strategies, particularly in high-risk groups like the Pima Indian community. Future research should consider more flexible statistical models to handle non-linear relationships and interactions among a broader set of predictors, enhancing the predictability and effectiveness of diabetes prevention programs.

References

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Centers for Disease Control and Prevention. (2023). *Type 2 diabetes*. https://www.cdc.gov/diabetes/basics/type2.html